

Veli — Frontend Design System

Cloud Parallax (Pvt) Ltd · v0.3 · 2026-05-02

1 Purpose

Canonical reference for visual and interaction patterns used in the Veli smartphone surface (Tamil-first PWA, Android wrap via Capacitor). Read before adding any UI. Pages reach for the primitives defined here rather than inline raw Tailwind classes for things the primitives already cover.

This is v0.2 — a civic-utilitarian system that replaces the prior shadcn-svelte zinc base. The shift was driven by a design exploration on claude.ai/design; the chosen aesthetic is “modern utilitarian / government-grade”, calibrated for rural Northern Province households on mid-range Android.

2 Four commitments

These are non-negotiable. Every design decision is judged against them.

Tamil	English	Detail
தமிழ் முதன்மை	Tamil-first	Designed in Tamil. English follows when it must, never as default. Display register: colloquial Jaffna Tamil, never literary, never Indian Tamil.
நம்பிக்கை	Trust by citation	Every claim links back to a government source or an audit trail. The platform never inflates a self-asserted artefact into something it is not.
குறை வளம்	Low-bandwidth	Works on a Rs. 18,000 Android with one bar of signal — or no signal at all. Service-worker offline cache is part of the contract.
மதிப்பு	Dignified	No condescension. Big targets, plain words, real receipts. No marketing language; no decorative SVGs.

3 Tokens

3.1 Colour palette

Three theme variants share the same token names; only the values change. The default theme is **civic**; **paper** is a warmer alternate; **ink** is the dark variant. Set on `<html>` via `data-theme`.

Token	Civic value	Use
--paper	#FFFFFF	Card surface.
--paper-2	#F7F4EE	Page background — warm off-white.
--paper-3	#EFEAE0	Inset / chip / panel.
--rule	#DCD5C7	Hairline borders, dividers.
--ink	#14140F	Primary text. 17.8:1 against --paper-2 (AAA).
--ink-2	#3B3A33	Secondary text.
--ink-3	#6B6859	Metadata, captions.
--primary	#0E4D3F	Civic green. CTAs, OK status, primary actions. 11.2:1 (AAA).

Token	Civic value	Use
--accent	#D9A441	Saffron-gold. Verification, official seals, hero dot. 3.1:1 (AA-large).
--alert	#A6363D	Deep red. Alerts, destructive actions, rejected. 6.8:1 (AAA).
--info	#2B4A6B	Ink slate. Information, secondary status. 9.4:1 (AAA).

Each non-neutral colour has a -soft companion (light tint, used as badge background) and an -ink companion (dark variant, used as badge text). Example: --accent-soft: #F8EDD3, --accent-ink: #6B4F14.

The full token set (including --rule-strong, --ink-mute, --primary-soft, --primary-ink, --alert-soft, --alert-ink, --info-soft) is declared in frontend/src/app.css. Tailwind v4 @theme inline exposes them as utility tokens (bg-paper-2, text-ink-3, bg-accent-soft, etc.).

Rule: never hex-code colours in templates. Always use the token, either as a CSS variable (color: var(--ink)) or a Tailwind utility (text-ink).

3.2 Typography

Three font families, each with a clear role:

Family	Token	Used for
Noto Sans Tamil	--f-tamil	Body, headings, all user-facing copy. Primary face. Weights 300–800.
Inter	--f-latin	Latin numerals, English transliterations, EN sub-labels.
JetBrains Mono	--f-mono	Section tags, metadata captions, IDs, receipts, SMS bodies, technical labels. Conveys “auditable / transparent”.

Type scale (display → caption):

Role	Size	Weight	Use
Display	72px	700	Hero only. The “வெளி.” mark on the landing page.
Heading 1	36px	600	Page titles. Section titles in the design-system page.
Heading 2	24px	600	Card titles, sub-section headings.
Title	18px	600	In-card titles, list-row titles.
Body	16px	400	Default copy.
Caption	13px	500	Helper text, hints, errors.
Mono · meta	11px	500	Section tags, metadata, IDs, technical labels.

3.3 Spacing and radius

Spacing scale (for components and layout): xs 4 · sm 8 · md 14 · lg 20 · xl 32 · 2xl 48 · 3xl 72. Density is variable: data-density on <html> toggles compact / regular / comfy, which scales the row heights and paddings of list rows and similar components.

Radius scale: xs 4 · sm 6 · md 10 · lg 14 · xl 20 · pill 999. Cards use `--r-md`; pills use `--r-pill`; inputs use `--r-sm`; large CTAs use `--r-md`.

3.4 Elevation

Three shadow tiers, very restrained — paper-feeling, never floating. `--sh-1` is a hairline. `--sh-2` adds a 1px y-offset. `--sh-3` is the raised state for popovers. Avoid heavier shadows.

3.5 Motion

Civic interfaces should not vibrate or shimmer. Motion is reserved for status changes, verification feedback, and transitions the user explicitly asked for. Five easings and five durations cover everything; **never** hard-code a cubic-bezier or a millisecond literal.

Easing	Curve	Use
<code>--ease-standard</code>	<code>cubic-bezier(.4,0,.2,1)</code>	Default — between two known places.
<code>--ease-entrance</code>	<code>cubic-bezier(0,0,.2,1)</code>	Things arriving · acknowledgements.
<code>--ease-exit</code>	<code>cubic-bezier(.4,0,1,1)</code>	Things leaving · dismissals.
<code>--ease-linear</code>	<code>linear</code>	Loading · indeterminate progress.
<code>--ease-spring</code>	<code>cubic-bezier(.34,1.56,.64,1)</code>	Receipts · success seal · saffron stamp.

Duration	ms	Use
<code>--dur-instant</code>	80	Micro feedback · press.
<code>--dur-fast</code>	140	Tooltip · chip selection.
<code>--dur-base</code>	220	Most transitions.
<code>--dur-slow</code>	360	Sheets · drawers · pages.
<code>--dur-ceremony</code>	680	Verification seal · receipt fold.

`prefers-reduced-motion`: reduce collapses every duration to 1ms (easings stay intact) and disables the skeleton shimmer + sync spinner animations. Critical state changes never depend on motion alone — every animated state has a static affordance.

4 Components

Every primitive lives at `frontend/src/lib/components/ui/<Name>.svelte` and is re-exported from the barrel `frontend/src/lib/components/ui/index.ts`. Pages import the primitive, never the underlying CSS classes when a primitive applies.

Primitive	Purpose
Button	Variants <code>primary</code> / <code>secondary</code> / <code>ghost</code> / <code>alert</code> ; sizes <code>sm</code> / <code>md</code> / <code>lg</code> ; block. 48 px min height (≥ 44 WCAG) at <code>md</code> . Composes the <code>.btn</code> class family.
Card + CardHeader + CardContent + CardFooter	Raised paper surface with hairline border, optional header / content / footer. Composes <code>.card</code> .
Badge	Inline pill for short generic state. Variants <code>primary</code> / <code>accent</code> / <code>alert</code> / <code>info</code> / <code>mute</code> . Optional dot.
TierBadge	Three-tier verification visual. Always renders both the EN and TA tier labels — never one without the other. Visual hierarchy follows legal effect: <code>self_asserted</code> → bronze, <code>community_corroborated</code> → silver, <code>authority_attested</code> → gold.

Primitive	Purpose
StatusStrip	Thin bar reporting backend reachability and phase number. Civic green when reachable; alert-soft when unreachable.
LocaleSwitcher	Three-button segmented control (தமிழ் / English / [?]?) calling Paraglide setLocale. Composes .seg.
Field	Label + input + (hint or error) wrapper. The <input>/<select>/<textarea> is supplied by the consumer with class="field-input".
LRow	The workhorse list row. Optional leading icon, title, meta, optional EN sub-label, optional trail (snippet). Used for procedure lists, document evidence, buyer offers, user fields.
Icon	Line icon (lucide-svelte under the hood). 24 px viewBox, 1.6 px stroke, currentColor. Filled variant available via filled prop.
OfflinePill	Saffron pill that reads “இணையமில்லை · OFFLINE”. Always-visible signal when the service worker is serving cached content.
Progress	Slim progress bar in civic green on a paper-3 track. Used for evidence completeness, sync progress.
Segmented	Generic segmented-control primitive (the same .seg used by LocaleSwitcher).
Trust	Citation card. Mark + title + body + optional ↳ cite. Used wherever a claim points to a government source or audit trail.
Receipt	Confirmation receipt with mono header, line items, and a centred receipt number. Designed to survive screenshots and printers.
SMS	Terminal-style SMS preview for the IVR/USSD/SMS fallback surface previews.
Phone	Mobile mockup frame for product-preview compositions in the docs.
Steps	Multi-step form indicator. 4–6 steps; each step has one decision. done / active / pending visual states; consumers pass { ta, en } per step.
StateCard	Empty / loading / error card with localized title + body and an optional footer snippet for an action button. The skeleton lines for loading state are passed in via children, not baked in.
Skeleton	Shimmer placeholder. lines={n} renders n staggered text-row skeletons; otherwise a single width × height block. Animation respects prefers-reduced-motion.
Stats + Stat	Counter grid for dashboards. Each Stat carries a value, optional inline unit, Tamil label, and optional English sub-label. 4-column on desktop, 2-column on mobile.
Timeline + TimelineRow	Vertical status timeline. Each row has a mono time, a dot (done / active / fail / pending), a Tamil title, and an optional desc.
SyncQueue + SyncRow	Offline sync queue display. Header with localized title + signal indicator; rows carry one of four states (queued / running / ok / error) with the appropriate glyph and animation.

4.1 TierBadge contract

The three tiers are normative. Their values, English labels, Tamil labels, and visual treatment are listed below. Never substitute, paraphrase, or invent a fourth.

Tier value	EN	TA	Visual
self_asserted	Self-asserted	சுய அறிவிப்பு	Bronze (tier-bronze) — visibly lowest-effort.
community_corroborated	Community-corroborated	சமூக சாட்சியம்	Silver (tier-silver) — neutral, no claim of legal effect.
authority_attested	Authority-attested	அதிகாரப்பூர்வ உறுதி	Gold (tier-gold) — saffron-soft tones, used only when the tier carries legal effect.

The dual-language label is the architectural contract from docs/plan/02_phase0_foundations.typ. A surface that needs to express a **derived** state (e.g. “expired attestation”) does so with a separate adjacent badge — never by changing the tier label.

5 Patterns

5.1 Hero

The landing page uses the giant Tamil wordmark with a saffron dot accent: `<h1 class="hero-mark">வெளி<span class="dot">.</span></h1>. clamp(72px, 12vw, 168px)` so it scales from phone to desktop. Bilingual metadata (`<dl class="hero-meta">`) sits beside or below the mark with `dt` in mono uppercase and `dd` in body.

5.2 Page shell

Every page wraps content in:

```
<div class="flex min-h-screen flex-col bg-background text-foreground">
  <main class="mx-auto w-full max-w-2xl px-6 py-12">
    <!-- page body -->
  </main>
</div>
```

Landing widens to `max-w-5xl`. User detail uses `max-w-2xl`. Avoid full-bleed on phones.

5.3 Section header

```
<div class="sec-tag">06 · கூறுகள் / Components</div>
<h2 class="text-[36px] font-semibold leading-tight tracking-tight">...</h2>
<p class="text-ink-2 max-w-[64ch]">...</p>
```

The mono section tag with the civic-green dot is the recurring section marker. Use bilingual `NN · தமிழ் /` English form when the section has a Tamil concept name.

5.4 List of properties

LRow rows inside a Card are the canonical way to display record fields (user details, document metadata, agricultural batch detail). Lead icon (40 px tile, paper-3 background) + title + meta + optional trail. Don’t use `<dl>` for record displays in v0.2 — LRow is the dignified pattern.

5.5 Trust / citation

Civic claims that come from a government source render as a Trust component (when added — currently in the design-system catalogue but not yet a primitive in this codebase). Until it lands, follow the markup

directly: `.trust` wrapper + `.trust-mark` (32 px primary circle with shield icon) + title + body + `.trust-cite` (mono caption prefixed with `↳`). Cite the source URL or the in-platform audit-trail reference, never an unverifiable claim.

5.6 Receipt

Confirmation numbers and submission receipts use the `Receipt` component (CSS class `.receipt`). Mono font, dashed top/bottom borders, the receipt-num as the centred final line. The user can screenshot it; SMS fallback delivers the same number.

5.7 SMS / USSD fallback

When bandwidth dies, surfaces fall back to SMS over the Veli-controlled shortcode (Ideamart). The SMS component shows the message as it will appear on a feature phone. Mono everywhere; preserves whitespace.

5.8 Form inputs

Every input is wrapped in a `Field` primitive: label above (never placeholder-only), `.field-input` for the control (full-width on mobile, 48 px tall, 1.5 px border), helper text below in `.field-help`, errors in `.field-error` (with the auto-rendered `!` icon). Required fields show a red `*` in the label.

5.9 Empty + error states

Localised, never English-only. Prefer the `StateCard` primitive over a generic `Card` for empty / loading / error states:

```
<StateCard kind="empty" title={m.user_not_found_title()}  
  body={m.user_not_found_body()}>  
  {#snippet children()}<Button kind="primary" size="sm">...</Button>{/snippet}  
</StateCard>
```

Loading state shows skeleton lines via `<Skeleton lines={3} />` — they should match the real layout that’s loading, never generic spinners. Errors **name the office** and offer a concrete next step (“Call 0212 222 555” or “Try again in 10 minutes”), never a stack trace and never a generic “something went wrong”. Never expose raw backend error envelopes to the user; log them.

5.10 Multi-step forms

Long government forms split across 4–6 screens, one decision per screen. Use the `Steps` primitive at the top to show progress; `active` is the current 0-based index. Each step has a Tamil `ta` label and an optional `en` sub-line in mono.

Local-first storage is the contract: every step’s data persists to IndexedDB before the network is touched. The `SyncQueue` primitive shows the user what’s pending: queued (`q`), uploading (`r`), ok (`k`), or errored (`e`). The queue is editable — users can cancel or retry rows.

5.11 Status timelines

Every government action gets a `Timeline` view. Citizens can subscribe to a request and see exactly which row is `active`. Past rows are `done` (civic green); failed steps are `fail` (alert red); future rows stay `pending` (default ring). Each row carries a mono time/relative-time on the left, a dot in the centre rail, and the Tamil title plus optional EN/meta description on the right.

6 Accessibility

A floor, not a ceiling. Six load-bearing rules; if a control fails any of them, it does not ship.

Rule	Detail
Contrast	Every text-on-background pair tested at 4.5:1 (body) or 3.0:1 ($\geq$ 24px or 16px-bold). Civic green on paper hits 11.2:1. Saffron is reserved for $\geq$ 24px or 16px-bold — never small body text. Dark mode swaps softs and inks together.

Rule	Detail
Targets	No tap target below 48×48 dp. Spacing tokens enforce this — buttons (48 px at md), list rows (- -d-row-h: 56px), and chips inherit a minimum height. Mid-range Android, sweat-on-glass, harvest gloves.
Input	No keyboard traps. IME-aware Tamil entry. Every form supports Tamil phonetic, English, and bilingual modes. Tab order matches reading order. Submit-on-enter is opt-in only.
Motion	<code>prefers-reduced-motion: reduce</code> is respected end-to-end. Saffron seal stamps, sheet animations, and skeleton shimmers fall back to instant cross-fades. Critical state changes never depend on motion alone.
Audio	Critical confirmations carry an optional Tamil voice line plus a single haptic pulse (<code>navigator.vibrate(40)</code>). Off by default in noisy/quiet zones. <code>aria-live="polite"</code> for status updates.
Literacy	Plain Tamil. Every step in a government flow has a recorded (not synthetic) voice-over button. Forms never depend on a citizen reading anything longer than 8 words.

Implementation specifics:

- **Focus rings** never disable. The - -ring token is - -primary; a 2 px ring at 2 px offset is the default focus halo.
- **aria-label** every interactive control needs one, **and it must be localised through Paraglide**. Never hard-code English.
- **Language tag** `<html lang>` is set by Paraglide via the `%paraglide.lang%` placeholder. Don't override.

7 Internationalisation

- **Tamil first.** Author the Tamil string before any other language. Never machine-translate Tamil from English in user-facing copy.
- **Register.** Northern Province colloquial (Jaffna). Native review is in the loop and authoritative.
- **Catalogue.** All keys in `frontend/messages/{en,ta,si}.json`, one key per UI string. Pluralisation and parameter substitution use Paraglide's parameter syntax — no string concatenation across keys.
- **Numerals.** Use European numerals (0–9) by default; Tamil numerals only when explicitly requested by content.

8 Voice and copy

Civic interfaces fail when they sound like government. Veli writes the way a careful neighbour would explain something: short sentences, real numbers, no jargon. Every error names the next action.

Sample tone (write / avoid):

Context	Write	Avoid
Confirmation	பிறப்பு சான்றிதழ் தயாராகி உள்ளது. Your birth certificate is ready.	Your requested document has been successfully processed.
Error	இந்தத் தொலைபேசி எண் வேறொருவரால் பயன்படுத்தப்படுகிறது. சரிசெய்ய அலுவலகத்தை அழைக்கவும் — 0212 222 555. That number is already in use. Call the office to fix it — 0212 222 555.	Error 0x4F2: Invalid input.

Context	Write	Avoid
Wait	அலுவலர் இப்போது பார்க்கிறார். வழக்கமாக இது 2 நாட்கள் ஆகும். An officer is reviewing this now. It usually takes 2 days.	Your request is queued.
Empty	இங்கே இன்னும் எதுவும் இல்லை. முதல் நிலத்தைச் சேர்க்கவும். Nothing here yet. Add your first land.	No record found.

Lexicon — words we standardise across every product:

Concept	Tamil	English fallback	Note
Submit	சமர்ப்பிக்க	Submit	Not “Send” — submission implies record.
Verify	சரிபார்	Verify	Pairs with the saffron seal.
Officer	அலுவலர்	Officer	Always name the office, not “the system”.
Receipt	பெறுசீட்டு	Receipt	Mandatory after every action that costs money or creates a record.
Pending	நிலுவையில்	Pending	Always pair with the working-day count.
Sync	பதிவேற்று	Upload	Tamil avoids “sync” — frame as “upload when signal returns”.

9 Mobile-first defaults

- Default breakpoints follow Tailwind: sm 640 px, md 768 px, lg 1024 px. The smartphone surface targets < sm; widen up from there.
- Never assume hover. Every interactive state must work on touch.
- Service worker caches the active session; pages render meaningful content with cached data plus a small banner when offline. Avoid spinner-only offline states.
- Capacitor Android wrap: status-bar colour is bg-paper-2; splash hidden on layout mount.

10 Code conventions

- **Svelte 5 runes only.** \$state, \$derived, \$effect, \$props.
- **Server-only** code lives in src/lib/server/. Never import in a .svelte component.
- **API calls** go through +page.server.ts load() (or form actions), using the typed client at \$lib/api. Never direct browser fetch.
- **Primitive imports.** Reach for \$lib/components/ui/<Name>.svelte before writing raw Tailwind for buttons, cards, badges, list rows, tier badges, status strips, fields, locale switchers, segmented controls, progress bars, offline pills, or icons.
- **cn() helper.** Compose classes with cn(...) from \$lib/utils.
- **Props.** Type with TypeScript interfaces; default values via destructuring at the \$props() site.
- **No inline <style>** unless scoped and unavoidable. Tailwind utilities first; CSS classes from app.css for composed components; CSS variables when a token is needed.

11 Boundaries

This design system covers the smartphone surface (PWA + Capacitor Android wrap). It does **not** cover:

- The IVR / USSD / SMS surfaces (governed by Tamil voice-prompt authoring rules, see strategic plan \$Phase 0 architecture).
- Internal admin or back-office tooling (out of scope for v1.x).

- Print or PDF artefacts (deed-ready bundles in Phase 2 use a separate Typst template).

When in doubt, defer to `CLAUDE.md` for project-wide rules and the strategic plan in `docs/plan/` for product policy.